

- 2 A light helical spring is suspended vertically from a fixed point, as shown in Fig. 2.1.

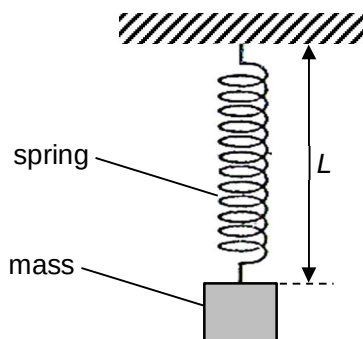


Fig. 2.1

Different masses are suspended from the spring. The weight W of the mass and the length L of the spring are recorded.

The variation with weight W of the length L is shown in Fig. 2.2.

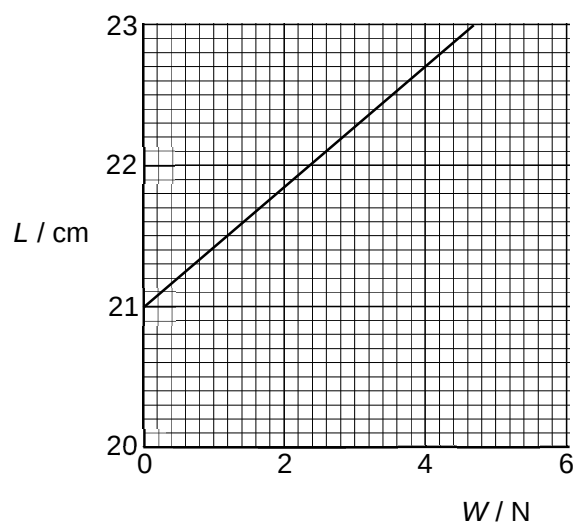


Fig. 2.2

- (a) A mass of weight 2.6 N is suspended from the spring.

When the mass is stationary, it is then pulled downwards a distance of 0.60 cm and held stationary.

For the increase in extension of 0.60 cm, determine the magnitude of the change in

- (i) the gravitational potential energy of the mass,

change = J [1]

(ii) the elastic potential energy of the spring.

change = J [2]

(b) The mass in (a) is now released. The mass performs simple harmonic motion.

(i) Use your answers in (a) to show that the total energy of oscillation of the mass is 0.0042 J.

[1]

(ii) Calculate the angular frequency of the oscillation.

angular frequency = rad s^{-1} [2]

- (c) A light card is attached to the mass, as shown in Fig. 2.3. The mass is pulled downwards a distance of 0.60 cm, and released at $t = 0$.

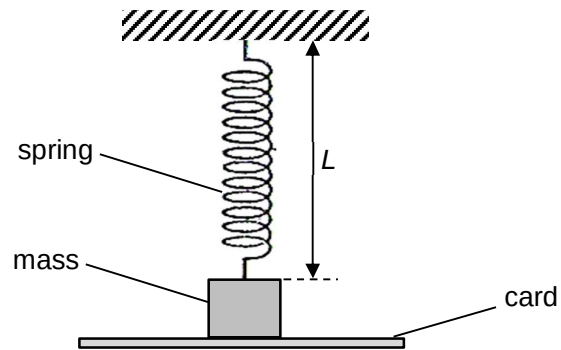


Fig. 2.3

The length of the spring is 21.7 cm after the first half of an oscillation.

Calculate the magnitude of the average force due to air resistance on the oscillating system.

force = N [3]

3 Two sources of sound S_1 and S_2 emit waves of frequency 220 Hz in phase as shown in